

Problem Set 6

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```
# Import modules
library(tidyverse)
```

Warning: package 'ggplot2' was built under R version 4.5.2

Warning: package 'tibble' was built under R version 4.5.2

Warning: package 'tidyr' was built under R version 4.5.2

Warning: package 'readr' was built under R version 4.5.2

Warning: package 'purrr' was built under R version 4.5.2

Warning: package 'dplyr' was built under R version 4.5.2

Warning: package 'lubridate' was built under R version 4.5.2

```
-- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
```

```
v dplyr      1.2.1      v readr      2.2.0
```

```
v forcats   1.0.1      v stringr    1.6.0
```

```
v ggplot2   4.0.3      v tibble     3.3.1
```

```
v lubridate 1.9.5      v tidyr      1.3.2
```

```
v purrr     1.2.2
```

```
-- Conflicts ----- tidyverse_conflicts() --
```

```
x dplyr::filter() masks stats::filter()
```

```
x dplyr::lag()     masks stats::lag()
```

```
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become
```

```
library(haven)

# Read the GSS data
gss <- read_dta("~/Documents/PAI741/Problem Set 6/Data/gss7224_r1.dta")
```

Recoding for Replication

```
# spkath, colath, libath

#
# Take the gss dataset and save changes back into gss
gss <- gss|>
  # Modify the spkath variable
  mutate(spkath = case_when(
    # If spkath equals 1, change it to 0
    spkath == 1 ~ 0,
    # If spkath equals 2, change it to 1
    spkath == 2 ~ 1,
    # All other responses are coded as missing
    TRUE ~ NA
  ))

# Now Repeat for other variables in Score

gss <- gss|>
  mutate(colath = case_when(
    colath == 4 ~ 0,
    colath == 5 ~ 1,
    # The rest is missing
    TRUE ~ NA
  ))

gss <- gss|>
  mutate(libath = case_when(
    libath == 1 ~ 0,
    libath == 2 ~ 1,
    # The rest is missing
    TRUE ~ NA
  ))
```

```

# Racists
# spkrac, colrac, librac
gss <- gss|>
  mutate(spkrac = case_when(
    spkrac == 1 ~ 0,
    spkrac == 2 ~ 1,
    # The rest is missing
    TRUE ~ NA
  ))

gss <- gss|>
  mutate(colrac = case_when(
    colrac == 4 ~ 0,
    colrac == 5 ~ 1,
    # The rest is missing
    TRUE ~ NA
  ))

gss <- gss|>
  mutate(librac = case_when(
    librac == 1 ~ 0,
    librac == 2 ~ 1,
    # The rest is missing
    TRUE ~ NA
  ))

# Communists
# spkcom, colcom, libcom
gss <- gss|>
  mutate(spkcom = case_when(
    spkcom == 1 ~ 0,
    spkcom == 2 ~ 1,
    # The rest is missing
    TRUE ~ NA
  ))

gss <- gss|>
  mutate(colcom = case_when(
    colcom == 4 ~ 0,
    colcom == 5 ~ 1,
    # The rest is missing

```

```

    TRUE ~ NA
  ))

gss <- gss|>
  mutate(libcom = case_when(
    libcom == 1 ~ 0,
    libcom == 2 ~ 1,
    # The rest is missing
    TRUE ~ NA
  ))

# militarists
# spkmil, colmil, libmil
gss <- gss|>
  mutate(spkmil = case_when(
    spkmil == 1 ~ 0,
    spkmil == 2 ~ 1,
    # The rest is missing
    TRUE ~ NA
  ))

gss <- gss|>
  mutate(colmil = case_when(
    colmil == 4 ~ 0,
    colmil == 5 ~ 1,
    # The rest is missing
    TRUE ~ NA
  ))

gss <- gss|>
  mutate(libmil = case_when(
    libmil == 1 ~ 0,
    libmil == 2 ~ 1,
    # The rest is missing
    TRUE ~ NA
  ))

# Homosexuals
# spkhomo, colhomo, libhomo
gss <- gss|>

```

```

mutate(spkhomo = case_when(
  spkhomo == 1 ~ 0,
  spkhomo == 2 ~ 1,
  # The rest is missing
  TRUE ~ NA
))

gss <- gss|>
mutate(colhomo = case_when(
  colhomo == 4 ~ 0,
  colhomo == 5 ~ 1,
  # The rest is missing
  TRUE ~ NA
))

gss <- gss|>
mutate(libhomo = case_when(
  libhomo == 1 ~ 0,
  libhomo == 2 ~ 1,
  # The rest is missing
  TRUE ~ NA
))

```

Replicate Figure 1

```

# Replicate figures
# Figure 1: 0-15 GSS Modified Stouffer Tolerance Battery, 1976-2024

# Create intolerance score by adding together the sum of all the values for the respondent.

# Take the gss dataset and save the changes back into gss (|> = then)
gss <- gss |>
  # Create a new variable called intolerance_score
  mutate(intolerance_score = spkath + colath + libath +
    spkrac + colrac + librac +
    spkcom + colcom + libcom +
    spkmil + colmil + libmil +
    spkhomo + colhomo + libhomo)

# Take the mean score of all respondents by year

```

```

# Take gss dataset
figure1 <- gss |>
  # group the observations by year
  group_by(year) |>
  # For each year, calculate the average intolerance score
  # na.rm = TRUE tells R to ignore missing values when calculating the mean
  summarize(mean_intolerance = mean(intolerance_score, na.rm = TRUE))

# Take figure1 and filter
figure1 <- figure1 |>
  # Keep only rows where mean_intolerance is not NaN
  filter(!is.nan(mean_intolerance))

# Show table of data for figure1
figure1

```

```

# A tibble: 26 x 2
  year      mean_intolerance
  <dbl>+<lbl>      <dbl>
1 1976            7.55
2 1977            7.65
3 1980            7.45
4 1982            7.41
5 1984            7.46
6 1985            7.41
7 1987            7.44
8 1988            7.44
9 1989            7.28
10 1990           7.20
# i 16 more rows

```

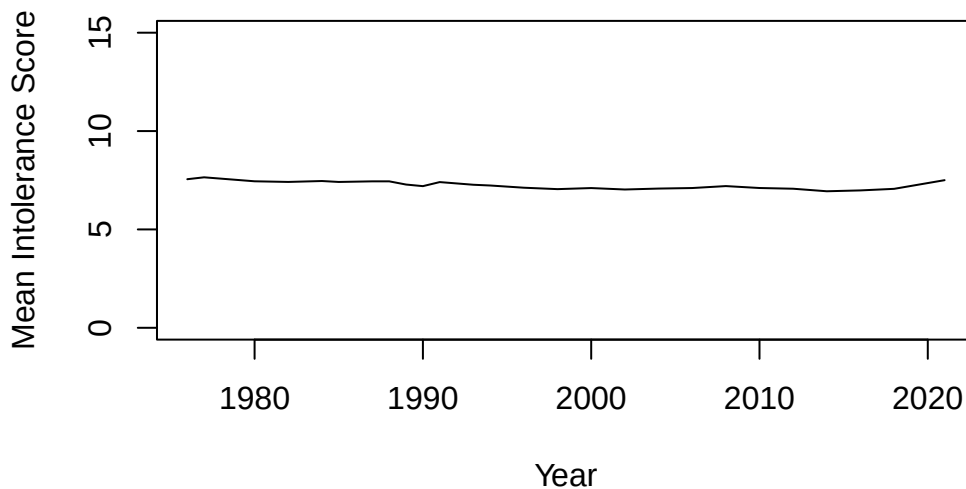
```

# Create plot 1
plot(figure1$year, figure1$mean_intolerance,
  # Setting the plot type to a line plot
  type = "l",
  # Main title of figure
  main = "The 0-15 GSS Modified Stouffer Tolerance Battery, 1976-2024",
  # X-axis label
  xlab = "Year",
  # Y-axis label
  ylab = "Mean Intolerance Score",

```

```
# Parameters for y-axis
ylim = c(0,15)
)
```

The 0–15 GSS Modified Stouffer Tolerance Battery, 1976–20



Replicate Figure 2

```
# Figure 2: Intolerance Towards Racists and Homosexuals, 1976–2024

# Take the gss dataset and save the changes back into gss
gss <- gss |>
# Create two new variables
mutate(
  # Create a racist_score by adding the three intolerance measures
  racist_score = spkrac + colrac + librac,
  # Create a homosexual_score by adding the three intolerance measures
  homosexual_score = spkhomo + colhomo + libhomo
)

figure2 <- gss |>
  group_by(year) |>
```

```

# Calculate the average racist and homosexual intolerance scores for each year
summarize(
  # Calculate the mean racist score for each year, ignoring missing values
  racist_mean = mean(racist_score, na.rm = TRUE),
  # Calculate the mean homosexual score for each year, ignoring missing values
  homosexual_mean = mean(homosexual_score, na.rm = TRUE)
)

# Remove years without data
# Take the figure2 dataset and save the filtered results back into figure2
figure2 <- figure2 |>
  # Keep only rows where both mean scores are not NaN
  filter(!is.nan(racist_mean) & !is.nan(homosexual_mean))

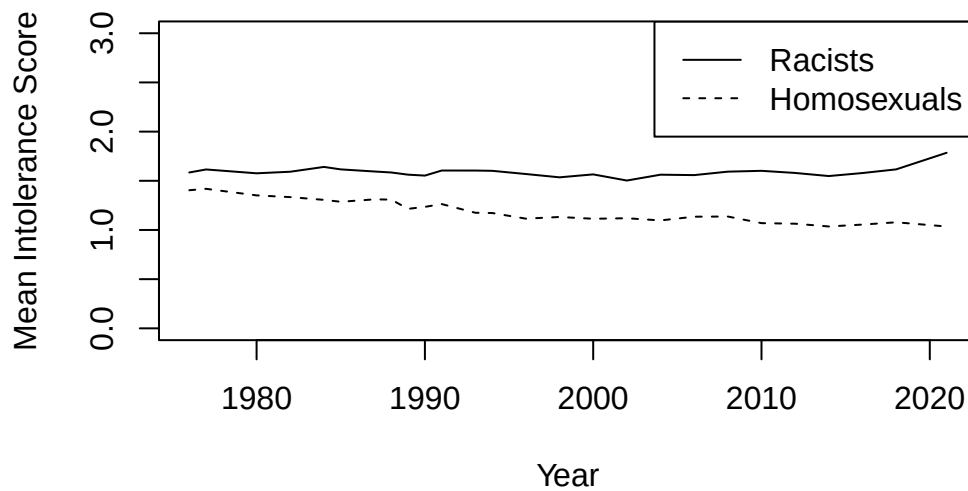
# Create plot
plot(figure2$year, figure2$racist_mean,
     type = "l",
     ylim = c(0, 3),
     main = "Intolerance Towards Racists and Homosexuals, 1976-2024",
     xlab = "Year",
     ylab = "Mean Intolerance Score")

# Add homosexual intolerance line
lines(figure2$year, figure2$homosexual_mean,
      lty = 2)

# Add legend
legend("topright",
      legend = c("Racists", "Homosexuals"),
      lty = c(1, 2))

```


Intolerance Towards Racists and Homosexuals, 1976–2024



Replicate Figure 5

```
# Figure 5: A Dichotomous Measure of Tolerance and Intolerance, 1976-2024
# If a respondent gives at least one intolerant response among the 15 items, they are classified as intolerant

# Create dichotomous measure

gss <- gss |>
  mutate(intolerant = case_when(
    intolerance_score == 0 ~ 0,
    intolerance_score > 0 ~ 1,
    # If there is a missing value for intolerance_score, keep it missing
    TRUE ~ NA
  ))

# Calculate proportion intolerant by year
figure5 <- gss |>
  group_by(year) |>
  # Calculate the proportion of respondents who are intolerant for each year
  # Ignore missing values when calculating the mean
```

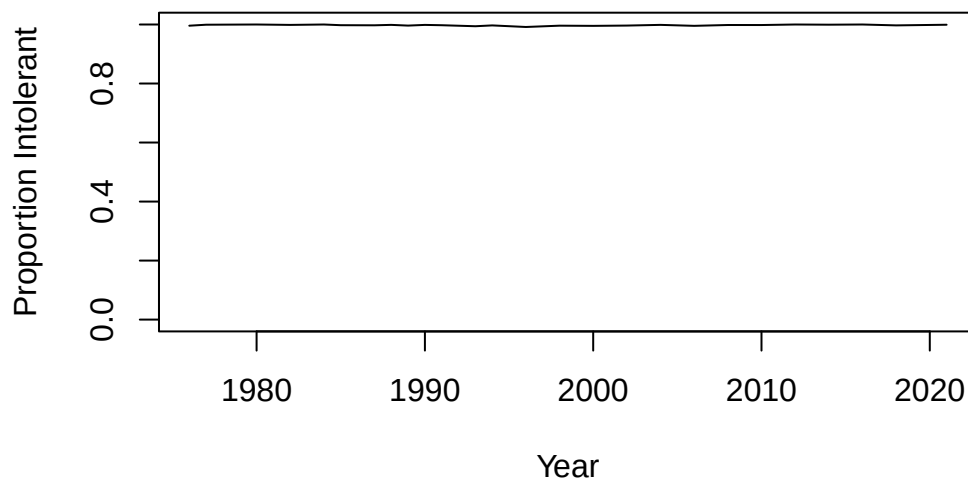
```

summarize(
  proportion_intolerant = mean(intolerant, na.rm = TRUE)
) |>
# Remove years with no data
filter(!is.na(proportion_intolerant))

# Create plot
plot(figure5$year, figure5$proportion_intolerant,
     type = "l",
     ylim = c(0, 1),
     main = "A Dichotomous Measure of Tolerance and Intolerance, 1976-2024",
     xlab = "Year",
     ylab = "Proportion Intolerant")

```

A Dichotomous Measure of Tolerance and Intolerance, 1976-



Original Work

```

# HEALTH

gss <- gss |>
mutate(health = case_when(

```

```

health == 1 ~ 4,
health == 2 ~ 3,
health == 3 ~ 2,
health == 4 ~ 1,
# The rest is missing
TRUE ~ NA_real_
))

# Central tendency and spread

# Calculate the average self-reported health score, ignoring missing values
mean(gss$health, na.rm = TRUE)

```

```
[1] 2.971599
```

```

# Calculate the standard deviation of self-reported health scores, ignoring missing values
sd(gss$health, na.rm = TRUE)

```

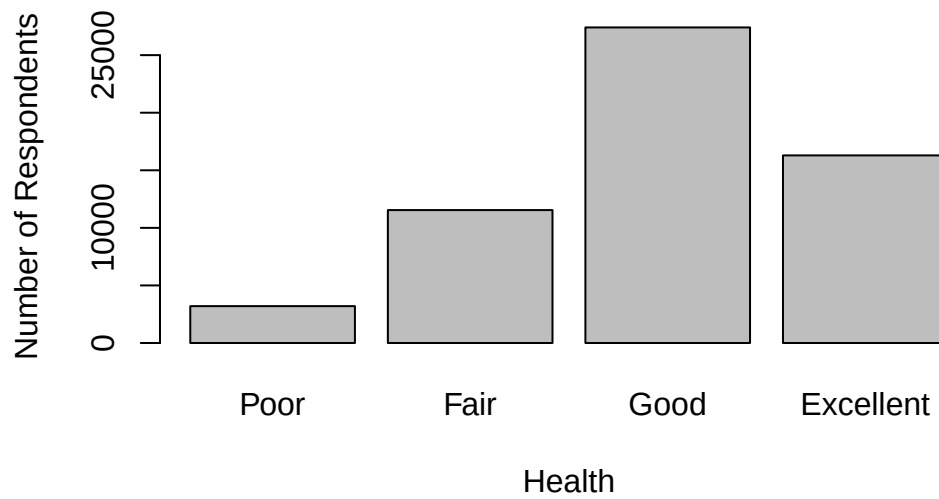
```
[1] 0.8335249
```

```

# Distribution of Health Responses
barplot(table(gss$health),
        main = "Distribution of Self-Reported Health",
        xlab = "Health",
        ylab = "Number of Respondents",
        names.arg = c("Poor", "Fair", "Good", "Excellent"))

```

Distribution of Self-Reported Health



```
# HAPPINESS

gss <- gss |>
  mutate(happy = case_when(
    happy == 1 ~ 3,
    happy == 2 ~ 2,
    happy == 3 ~ 1,
    # The rest is missing
    TRUE ~ NA_real_
  ))

# Central tendency and spread
mean(gss$happy, na.rm = TRUE)
```

```
[1] 2.154849
```

```
sd(gss$happy, na.rm = TRUE)
```

```
[1] 0.6448012
```

```
# Distribution of Happiness Responses
barplot(table(gss$happy),
        main = "Distribution of Self-Reported Happiness",
        xlab = "Happiness",
        ylab = "Number of Respondents",
        names.arg = c("Not Too Happy", "Pretty Happy", "Very Happy"))
```



Where did you encounter problems with this replication? Was there anything that was particularly tricky?

I had a difficult time identifying which variables to use in the code book to replicate the figures, as there were over 6,000 columns.

How would you interpret (in other words, explain to a general audience) what Mondak and Sanders found in figures 1, 2, and 5?

Figure 1 Interpretation: According to Mondak and Sanders, the Americans became more tolerant. Figure 1 uses the 0-15 GSS tolerance scale, where a higher score indicates greater intolerance. The average score fell from 7.24 in 1976 to 5.28 in 1998. However, Mondak and Sanders argue that people may simply have become less hostile towards some of the groups included in the survey.

Figure 2: Figure 2 helps explain why the authors are skeptical of Figure 1. They separate responses concerning racists and homosexuals and find very different trends. Intolerance toward homosexuals declined by more than intolerance toward racists.

Figure 5: Figure 5 suggests the change in tolerance is much smaller. Instead of asking how many intolerant responses someone gives, they divide respondents into two categories: someone who gives zero intolerant responses is classified as tolerant, while someone who gives at least one intolerant response is classified as intolerant. The proportion classified as intolerant falls only slightly from about 85% in 1976 and 1988 to 79% in 1998, a decline of less than six percentage points.

Has anything changed since 1998 that would alter the substantive takeaway points in their findings?

My updated figures suggest that Mondak and Sander's main conclusion still holds. My extension shows that from roughly 1998 onward, the overall intolerance score does not continue falling. Instead, it fluctuates around roughly 8 in my replication and even rises somewhat in the most recent observations. So there is little evidence in my extended series of a continuing long-term movement toward greater tolerance after 1998. For figure 2, my extension shows that attitudes towards the two groups continue to move differently rather than converging around a common trend. This reinforces the idea that a singular tolerance score can mask substantial differences in attitudes towards a specific group. Lastly, figure 5 (continues) shows that the proportion remains pretty high and flat, suggesting that the targets of intolerance may have changes, but the proportion of intolerant respondents likely did not change substantially.